

Neuro-Evolved Heuristics for Variable Gapped Common Subsequence Identification

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Introduction

- Objects we deal with: sequences (strings) over finite alphabet Σ
 - DNA/RNA sequences over $\{A, T, G, C/U\}$
 - Proteins over 20 (canonical) amino acids: $\{A, C, D, E, P, Q, \dots\}$
- **One of central tasks in computational biology:**
 - sequence comparison, finding **common motifs** between sequences
 - compare structurally, relate it to functionality
- **Subsequences:** reveal structural similarities → **Longest common subsequence problem** variants (studied 50 years already)

Longest common subsequence problem (LCSP)

Definition (LCSP)

Input: Given an arbitrary set of sequences $S = \{s_1, \dots, s_m\}$

Task: Find a subsequence s **common** for all sequences from S with **maximum** possible length ($|s|$).

Example

Input: $S = \{\text{AATTGC}, \text{ATTAC}\}$

LCS solution: $s = \text{ATTC}$

LCS: Literature & Problem Variants

- For $m = 2$, polynomially solvable: **Dynamic programming**, Hunt-Szymanski, Hirschberg, ...
- When m arbitrary large – $\mathcal{NP}$ -hard:
 - approximation, meta-heuristic, but also exact approaches

Problem-related practical variants:

- Repetition-free, filled LCS problem, **Gapped LCS problem**

The gapped LCS problem (Peng and Yang, 2012)

Definition (A gap sequence)

Given is a sequence s and an assigned function $G_s: \{1, \dots, |s|\} \mapsto \mathbb{N}$. A pair (s, G_s) is called a gap sequence.

(To each position of seq. s , one number is assigned)

Definition (A gapped subsequence)

Sequence $\tilde{s}$ is a **gapped subsequence** of (s, G_s) iff

- $\tilde{s}$ is a **subsequence** of s
- the gap constraint G_s fulfilled resp. *positions of embedding* $\tilde{s}$ in s :
 - suppose $i_1 < \dots < i_{|\tilde{s}|}$ are those **positions**
 - $(\forall j = 2, \dots, |\tilde{s}|) \text{ } i_j - i_{j-1} \leq G_s(i_j) + 1$ (consecutive positions respect gap distances = allowed *number of letters inbetween*)

Problem definition

Example

$s = \text{AATTGC}$, $G_s(\cdot) = 1$

- $\tilde{s} = \text{ATG}$, the embedding: **AATTGC** (valid gapped subsequence)
- $\tilde{s} = \text{ATG}$, the embedding: **AATTGC** (invalid gapped subsequence)

Definition (The multiple (variable) gapped LCS problem – MVGLCSP)

Input: Given is a set of gapped sequences $\{(s_1, G_{s_1}), \dots, (s_m, G_{s_m})\}$.

Task: Find the longest sequence $\tilde{s}$ so that

- $\tilde{s}$ is **common subsequence** of each s_i
- $\tilde{s}$ **fulfills all gap constraints** G_{s_i} ($i = 1, \dots, m$)

Note: When $G_{s_i} = n$ (length of longest sequence) $\Rightarrow$ VGLCSP = LCSP.

Literature & Motivation for VGLCSP

- Peng and Yang (2012, 2014): studied $m = 2$ version by **three dynamic programming** (DP) approaches (basic and two advanced)
- Manea et al. (2024): complexity analysis investigated
 - **NP-hard** under arbitrary large m
- Djukanovic et al. (2026): the first approach for general VGLCSP (m arbitrary)

Motivation:

- **Genetics and molecular biology**: DNA/protein comparison analysis where structural distances between residues must be respected
- **Time-series analysis**: in settings where events are required to occur within specified temporal delays (Lainscsek et al. (2015))

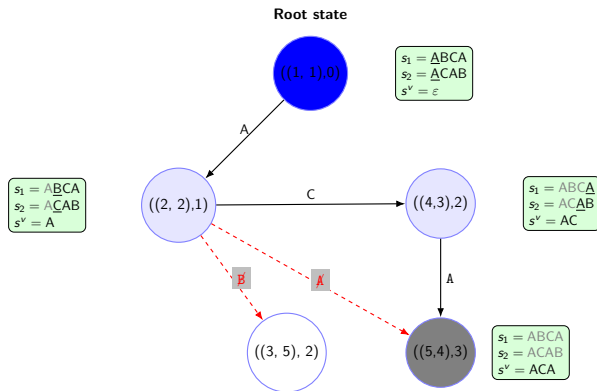
Methodology by Djukanovic et al. (2026)

- Based on the **state space graph formulation** of the problem
- **Nodes** corresp. to partial solutions and resp. subproblems be solved
- **Gap constraints:** incorporated to cut-off invalid extensions (edges) among the LCS extensions immediately
- **Many root/source nodes** in the state graph—the position of leading letter in each common gapped subsequence free to select
 - root state space subgraphs (formulation) w.r.t. starting point (match)...

Root-based state graph formulation: example

Given $s_1 = \boxed{A}BCA$, $s_2 = \boxed{A}CAB$, assuming $G_1 = G_2 = 1$, match $p^L = (1, 1)$;

SUBSPACE((1, 1)): partial solutions start with A at (1, 1)

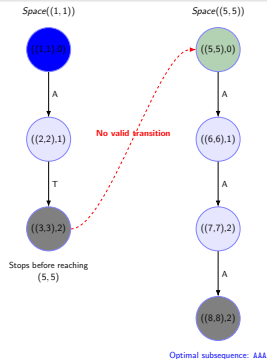


SUBSPACE((1, 1)) $\Rightarrow$ ACA

An issue with the root-state-space formulation: disconnected components

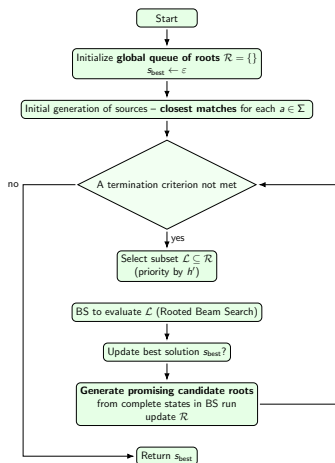
Example

$S = \{s_1 = \text{ATGG} \boxed{\text{A}} \text{AA}, s_2 = \text{ATCC} \boxed{\text{A}} \text{AA}\}$, with $G_{s_1} = G_{s_2} = 1$.



Observation: $((5,5), *) \notin \text{SUBSPACE}((1,1))$, The optimal solution: **AAA**.

Workflow of the Iterative Multi-source Beam Search – IMSBS



Heuristic guidances in IMSBS

Three LCS heuristic guidances to guide BS and prioritize root nodes most sensible:

- ① UB₂: *Character Frequency Alignment* score the best performing

Experimental evaluation – Conclusions:

- IMSBS able to efficiently run on large synthetic instances
- Obtain a large portion of known optimal solutions for $m = 2$
- **Weaknesses:** the performance of UB₂ seems to significantly drop for large n ($n = 500$) and small $|\Sigma|$ ($|\Sigma| = 2$)

⇒ **Construction of a new heuristic guidance**

Designing advanced heuristic guidances

TODO